

speed, distance, heart rate, and many more has dominated the lower-end of the market.

“You never know. The next generation of truly wireless might have 4G connectivity where you can just talk with them. Work is going on in building an MP3 player in earwear so that there is no need to carry your phone when you go outside,” says Amit Khatri.

It is also believed that the next generation of watches will have many smart features. So rather than buying new sets of watches, people can simply change the outer case, with the inner chip remaining intact. This will not only save expenditure but also retain the usefulness and power of the same chip. To achieve this, the OS must be small yet powerful enough to interact with everyone—be it industrial or home application devices.

“We are looking at smart apparels, muscle simulation technology, and devices that can gather respiratory and heart-rates,” says Pratik Saraogi, CEO, Actifit. “We are also looking at EMG and EMS sensors that simulate muscle groups during workouts. And these will come all under the category of wearables.”

Capattery, a next-gen super-fast battery charging company, is making solutions that allow a battery to be charged within a few seconds (in contrast to Li-Ion/LiPo battery that takes approximately thirty minutes to charge a wearable). This implies that a 5000mAh mobile phone can be charged in less than two minutes.

Power of gestures

Voice recognition has been there longer than gesture recognition. But it is hard to imagine people going around and speaking to their devices to perform certain actions.

Another challenge voice recognition has is the complexity of interpreting languages and dialects that vary from culture to culture. Gestures, however, seem to be more universal in that sense. For example, if you point your fingers upwards, it will always mean the same to everyone, that is, to go up—regardless of the language you speak.

“Gesture recognition is much more universal in its appeal and reach,” says Abhishek Satish. He further adds that there are certain challenges with

Speakers



Aashish Kumbhat, founder and director, Inbase



Abhishek Satish, co-founder and director, Vicara



Amit Khatri, co-founder, Noise



Naresh Neelakantan, senior architect-system software, Sasken Technologies



Nikhil Bhaskaran, founder, ShunyaOS



Pratik Saraogi, CEO, Actifit

respect to it as well since a sensor or an image processing device is required to be connected to your hand, which consumes a lot of power. At the end of the day, battery technology will determine how gestures are used for communication.

In short, AI-based gesture recognition devices will outweigh all other kinds in the future.

And as stated by Atul Khatri, if your design is not viable, then there is no point in selling as it will have no takers. One needs to make a technology that is commercially acceptable by consumers.

The Indian government, through its Make In India programme, is enabling a vision wherein goods are manufactured locally, leading to cost-effectiveness in production and thus saving enormous amounts of money. The government is also rolling out a PLI (production linked incentive) scheme, which is specifically targeted to smart wearables manufacturing. With that coming in, it will be a big boost for the Indian industry and help put them on a global pedestal.

Multiple design opportunities

Some of the points corresponding to both Indian and global spaces are:

1. Tackle various markets differently. For instance, setting the right price plays an important role in an Indian context in contrast to the USA where integration with other devices is preferred.

2. Be specific to the needs of the consumer. Indian consumption and usage patterns are different from other parts of the world.

3. People are more inclined towards the experience they receive from a product rather than its functionality. Go for the former.

In conclusion, the technological challenges with regards to hardware and software are growing. On top of that, consumer demands are also changing. All of this has made the wearable design a bit daunting. But it has also opened up new possibilities and new markets to discover. And thanks to the government initiatives, the obstacles will soon be surpassed. **EFY**

This article is based on the panel discussion ‘Design Challenges around Smart Wearables and Solutions’ that took place during January edition of Tech World Congress 2021. The panel was chaired by Naresh Neelakantan, senior architect-system software at Sasken Technologies.